The Age of the Universe

1) **<u>בראשית רבה (וילנא) פרשה ג</u>**

ז אר"י בר סימון יהי ערב אין כתיב כאן, אלא ויהי ערב, מכאן שהיה סדר זמנים קודם לכן, א"ר אבהו מלמד שהיה בורא עולמות ומחריבן, עד שברא את אלו, אמר דין הניין לי, יתהון לא הניין לי, א"ר פנחס טעמיה דר' אבהו וירא אלקים את כל אשר עשה והנה טוב מאד דין הניין לי, יתהון לא הניין לי.

2) **<u>בראשית פרק ז פסוק כג</u>**

וימח את כל היקום אשר על פני האדמה מאדם עד בהמה עד רמש ועד עוף השמים וימחו מן הארץ וישאר אך נח ואשר אתו בתבה׃

3) **<u>העמק דבר להנצי"ב, בראשית פרק ז פסוק כג</u>**

אבל נשתיירו כמה גופות שנפל עליהם עפר הרבה ע"י שטף המים ונשארו הגופות קיימין והן הנה הגופות שמוציאין החופרי ארץ ומוצאין עצמות מבריות שלא נמצא עתה בעולם ומזה שפטו הרבה שהי' לפני בריאה זו עולם אחר ואז היו בריות אחרות . . . מ"מ קשה לדעתי . . . היה מסתכל בהן ולא היו ערבים עליו ומחזירן לתהו ובהו וא"כ לא נשתיירו מהם שריד ופליט . . . ומה שמוצאין בריות משונות הוא ממה שהרכיבו שני מינים שונים ונולד ע"י זה בריות משונות

4) **<u>מלבי"ם בראשית פרשת נח פרק ז פסוק כג</u>**

מבאר וימחו מן הארץ, שהגם שהרבה נשארו עצמותיהם החזקים כמטילי ברזל ולא נמוחו, בכ"ז נמחו מן הארץ כי ע"י שטף המים הובלו הפגרים לתוך העמקים ורובם נבלעו בעמקי תהום, אשר האדמה פצתה את פיה מעומק תהום רבה וירדו כמה אלפים אמה לעמקי שאול, עד שבצאת נח מן התיבה לא מצא שום רושם מפגרי בע"ח ועצמות הענקים ובע"ח הגדולים שהיו קודם המבול. ועדות ה' נאמנה מחכימת פתי להשיב דבר לחכמי הגעאלאגיא, שחופרים בעמקי האדמה ומוצאים עצמות גדולות מענקים ובע"ח גדולים שנאבדו מן הארץ ואינם עוד מימי המבול ואילך, והם מוכיחים מזה קדמות העולם כי יחשבו את האדמה שעשויה מינים מינים, שבכל עומק ידוע יש מין אדמה אחרת, ויחשבו לפי איכות האדמה שימצאו העצמות שם, ולפי חשבון כמה היה צריך עד שיולדו שטחי האדמה שהעולם קדמון מרבוא שנים, ושכבר היו בע"ח מימי קדם קודם לזמן היצירה שלנו. וכ"ז הבל וריק, כי בעת המבול פתחה האדמה את לועה ע"י הרעשים שבאו מעומק שאול ותהום רבה והפכה תחתונים למעלה ועליונים למטה והורידה פגרי הבע"ח עמוק עמוק, ונבקה חכמת הדורשים והחופרים לדעת איכות המהפכה הזאת אשר עשתה נפלאות במשך ק"ן יום של עמידת המים, עד שכל הסדרים שסדרו למו החוקרים מאיכות האדמה בכל שטח וכל חשבונותיהם בזה נתבלבלו אז וכל השערותיהם ודמיונותיהם ישא רוח יקח הבל, ולו חכמו ישכילו זאת ויראו מעשי ה' ונפלאותיו במצולת שאול איך מחה את כל היקום מעל פני האדמה ויטביעם בשאול תחתית עדי אובד׃

5) **<u>Einei Yisroel (Rav Yisroel Belsky) Bereishis pp. 12-16</u>**

Nowadays, scientists have embraced a theoretical explanation of the origins of the universe called 'the Big Bang' However, what is understandable about cosmogony has already been discussed by the Ramban (ד"ה והנה1:1)
Based on kabbalah, the Ramban says that HaKadosh Baruch Hu initially created an infinitesimally small, non-dimensional speck (נקודה קטנה דקה ואין בה ממש), called "tohu". This speck contained all the potential, formless elements of Shamayim vaaretz, Heaven and earth. HaKadosh Baruch Hu further imbued the primordial speck with an elemental structure, called "bohu". Subsequently, Hashem brought forth the myriad of potential elements and structures of the universe into actual existence. Recent measurements taken of various elements in the universe indicate that the universe is expanding, and that has led even secular scholars to make conclusions similar to what the Ramban wrote over seven hundred years ago.
To the extent to which this theory matches the Ramban's description of the origins of the universe, can it aid in fostering a deeper understanding of certain aspects of Maaseh Bereishis. One must beware of bending the words of the Ramban to fit a particular scientific theory, however, as there are elements to the theory that have no basis whatsoever in Torah, and are therefore incorrect.
The above Ramban makes it clear that the entire known universe is essentially derived from a simple point of primordial matter, which implies that the whole of creation should share some basic elemental structure. Though smaller subatomic particles exist, atomic research has shown that all elements can be differentiated by the number of protons in their nucleus. The simplest is hydrogen, with only one proton. Other elements are simply multiples of this basic element. . .

More recently, astronomers have observed that one particular astronomical phenomenon produces these elements. Once in a great while, massive stars called Red Giants, thousands of times larger than our Sun, have been observed to collapse under their own weight as their internal furnaces run out of fuel. Called a supernova, the cata clysmic crumbling of the star creates a massive nuclear fireball that produces all types of atomic elements, including oxygen and carbon. To scientists, this discovery helped explain how the building blocks of life were formed. To us, it is a fantastic example of how the destruction of worlds formed the foundation of our own world's existence. Much of the worlds energy needs are provided by what we call "fossil fuels. These include oil, gasoline, natural gas and coal. The organic nature of these substances reveals that they are derived from the remains of plants and animals. These fossil fuels are extracted from deposits deep beneath the earth's surface. The vast quantity of this decayed organic material suggests that there once existed large forests of green leafy plants that captured the energy of the sun. The scientific world theorizes that this primordial world was destroyed by the impact of a large meteorite. Gradually, these organic remains were compressed into deep subterranean layers, which eventually fossilized and transformed into coal, oil, and gas. Whatever the cause, it is clear that the death and destruction of billions of tons of plant and animal life at some point in the past, became our greatest source of energy. Thus, we have yet another example of the dependency of our world on the destruction of a previous one. . .
The Ribbono Shel Olam built His palace on a heap of ruined worlds for a reason. Our primary source of energy could be the result of a lost world of lush greenery and giant creatures. The basic building blocks of life, carbon and oxygen, might be the products of the systematic and cataclysmic destruction of primordial suns. Even the existence of our universe is the outcome of the obliteration of the original speck of "Heaven and earth." Finally, the fertile ground that brought forth our world's vegetation, was likely the result of the pulverization of a desolate, uninhabitable and rocky world.

6) **מסכתות קטנות מסכת אבות דרבי נתן נוסחא א פרק לא**

רבי אליעזר בנו של רבי יוסי הגלילי אומר תשע מאות ושבעים וארבע דורות קודם שנברא העולם היתה תורה כתובה ומונחת בחיקו של הקב"ה ואומרת שירה עם מלאכי השרת שנאמר ואהיה אצלו אמון ואהיה שעשועים יום יום [משחקת לפניו] ואומר משחקת בתבל ארצו [ושעשעי את בני אדם] (משלי ה' ל' ול"א):

7) **תלמוד בבלי מסכת חגיגה דף יג עמוד ב**

תניא, אמר רבי שמעון החסיד: אלו תשע מאות ושבעים וארבע דורות שקומטו להיבראות קודם שנברא העולם ולא נבראו

8) **תלמוד בבלי מסכת סנהדרין דף צז עמוד א**

אמר רב קטינא: שית אלפי שני הוו עלמא וחד חרוב, שנאמר (ישעיהו ב') ונשגב ה' לבדו ביום ההוא. אביי אמר: תרי חרוב, שנאמר (הושע ו') יחיינו מימים ביום השלישי יקמנו ונחיה לפניו, תניא כותיה דרב קטינא: כשם שהשביעית משמטת שנה אחת לשבע שנים, כך העולם משמט אלף שנים לשבעת אלפים שנה, שנאמר ונשגב ה' לבדו ביום ההוא, ואומר (תהלים צ"ב): מזמור שיר ליום השבת - יום שכולו שבת. ואומר: (תהלים צ') כי אלף שנים בעיניך כיום אתמול כי יעבר. תנא דבי אליהו: ששת אלפים שנה הוי עלמא, שני אלפים תוהו, שני אלפים תורה, שני אלפים ימות המשיח.

9) **Rabbi Menachem M. Schneerson, A Letter on Science and Judaism, Challenge p. 145**

The argument from the discovery of fossils is by no means conclusive evidence of the great antiquity of the earth, for the following reasons:
(a) In view of the unknown conditions which existed in "prehistoric" times, conditions of atmospheric pressures, temperatures, radioactivity, unknown catalyzers, etc., etc. as already mentioned, conditions that is, which could have caused reactions and changes of an entirely different nature and tempo from those known under the present-day orderly processes of nature, one cannot exclude the possibility that dinosaurs existed 5722 years ago, and became fossilized under terrific natural cataclysms in the course of a few years rather than in millions of years, since we have no conceivable measurements or criteria of calculations under those unknown conditions.
(b) Even assuming that the period of time which the Torah allows for the age of the world is definitely too short for fossilization (although I do not see how one can be so categorical), we can still readily accept the possibility that God created

ready fossils, bones or skeletons (for reasons best known to Him), just as He could create ready living organisms, a complete man, and such ready products as oil, coal, or diamonds, without any evolutionary process.

10) **אליהו זוטא (איש שלום) פרשה ג**

הכל ברא הקב"ה בעולמו חוץ ממידת השקר שלא ברא

11) **Rabbi Shimon Schwab, How old is the Universe?, Challange p. 165**

However, nothing compels us to assume that *before* the Sabbath, while the Creator was still engaged in the process of creating, this synchronization had already taken place. During the period of creation, while the time-system of the Hidden Light - the "cosmic time clock" was functioning regularly as it does today, the earth could have turned around its axis much more rapidly. This would mean, that while the creation Light would register only one day, the earth could have experienced any number of days.
Following this method of reasoning to the very end, we can now advance the following suggestion, that a) the *duration* of the six regular earth days and b) that the billions of years which science claims to have calculated, all actually occurred during six ordinary days. Even if science could ever substantiate its theory of longevity, it could never be construed as contradiction to the Torah. *Billions of years during the era of creation are equal to six regular days today.*

12) **תלמוד בבלי מסכת סנהדרין דף לח עמוד ב**

אמר רבי יוחנן בר חנינא: שתים עשרה שעות הוי היום; שעה ראשונה - הוצבר עפרו, שניה - נעשה גולם, שלישית - נמתחו אבריו, רביעית - נזרקה בו נשמה, חמישית - עמד על רגליו, ששית - קרא שמות, שביעית נזדווגה לו חוה, שמינית - עלו למטה שנים וירדו ארבעה, תשיעית - נצטווה שלא לאכול מן האילן, עשירית - סרח, אחת עשרה - נידון, שתים עשרה - נטרד והלך לו.

13) **תפארת ישראל, דרוש אור החיים** (Translation Printed in **Challenge**)

We are enabled to appreciate to the full the wonderful accuracy of our holy Torah when we see that this secret doctrine, handed down by word of mouth for so long, and revealed to us by the sages of the Kabbala many centuries ago, has been borne out in the clearest possible manner by the science of our generation.
The questing spirit of man, probing and delving into the recesses of the earth, in the Pyrenees, the Carpathians, the Rocky Mountains in America, and the Himalayas, has found them to be formed of mighty layers of rock. lying upon one another in amazing and chaotic formations, explicable only in terms of revolutionary transformations of the earth's surface.
Probing still further, deep below the earth's surface, geologists have found four distinct layers of rock, and between the layers fossilized remains of creatures; those in the lower layers being of monstrous size and structure, while those in the higher and more recent layers being progressively smaller in size but incomparably more refined in structure and form.
Further, they found in Siberia in 1807, under the eternal ice of those regions, a monstrous species of elephant, some three or four times larger than those found at the present day; the skeleton may still be seen in the Zoological Museum at St. Petersburg. Since that icy region is incapable of supporting any species of elephant, we must conclude either that the creature was swept there as a result of some cosmic upheaval, or that in some previous epoch the climate of Siberia had been warm enough to support elephants.
Similarly, fossilized remains of sea creatures have been found within the recesses of the highest mountains, and scientists have calculated that of every 78 species found in the earth, 48 are species which are no longer found in our present epoch.
We know, too, of the remains of a giant creature discovered deep in the earth near Baltimore, 17 ft. long and 11 ft. high; specimens of which have also been found in Europe, and which has been given the name 'mammoth'.
Another giant creature whose fossilized remains have been found is the iguanadon, which stood 15 ft. high and

whose length extended to 90 ft; from its internal structure scientists have concluded that it was herbivorous.
Another is the megalosaurus, which was slightly smaller than the iguanadon, but which was carnivorous.
From all this, we can see that all the Kabbalists have told us for so many centuries about the fourfold destruction and renewal of the earth has found its clearest possible confirmation in our time.
And if we know how to look, we shall see that the Holy Torah itself hints at these facts in its opening verses. The first verse refers to the original act of creation, while "the earth was void and waste" refers to the epochs of upheaval and destruction which preceded our present age. The Torah passes over the intervening epochs in silence, because they have no immediate relevance to us; but "the spirit of God" which "was moving over the face of the waters" denotes the spirit of life which was, so to speak, waiting to reenter the creation to be the vehicle of the glory of God. The ascending scale of being of the seven days of creation, then reflects the ascending scale of the great cosmic cycle.
The very first letter of the Torah, with its traditional four *taggin* hints that our present epoch is the fourth, and the fact that it is a *beth,* and written large, indicates that the greatest peak of creation, a thinking being, now inhabits the world for the second time.
For, in my opinion, the prehistoric men whose remains have been discovered in our time, and who lived long before Adam, are identical with the 974 pre-Adamite generations referred to in the Talmud (Shabbat 88 and Hagiga 14), and lived in the epoch immediately before our own.

14) **THE AGE OF THE UNIVERSE A Torah-True Perspective By Rabbi Aryeh Kaplan February 18, 1979**

One of the most recent gedolim to speak about this shitah is the author of Tiferes Yisroel, at the end of Seder Nezikin, in what he called Drush Ohr HaChaim. And I might say that this is very controversial in many quarters. Most people assume that it was written as apologetics to Darwin's theories, not realizing that it was first published in 1834, long before Darwin came on the scene. This Drush was so controversial, that it is omitted in many editions of the Yachin U'Boaz Mishnayos, which normally contains it. On more than one occasion, I wanted to show someone something in this Drush, only to find that it had been ripped out of the book! This is also why some Chassidic groups do not learn Tiferes Yisroel, even though he was a great gadol.

15) **עולמות שחרבו (פרקים נבחרים מחייו ופעליו של כ"ק אדמו"ר הגאון הקדוש איש האלקים ראש גולת אריאל מרן יואל טייטלבוים זצוקללה"ה אדמו"ר מסאטמאר) חלק ראשון ע' קנ"ז**

פעם סיפרתי לו כי גדול אחד התמרמר לפני על שום מה מדפיסים בא"י את המשניות עם הדרוש אור החיים שאין רוח חכמים נוחה מזה ולמה לא משמיטים דרוש זה מרן זיע"א לא הגיב על סיפור זה (בניגוד לדרכו) הבנתי מזה כי מחמת כבודו של מרן תפארת ישראל כבש דבריו ולא הגיב דבר ולא הוציא אך מלה

16) **אוצר החיים** (Translated by Rabbi Aryeh Kaplan in **Immortality, Resurrection and the Age of the Universe, A Kabbalistic View**)

I, the insignificant Isaac of Akko, have seen fit to record a great mystery that should be kept very well hidden. One of God's days is one thousand years, as it is written, "For a thousand years in Your sight are as a day" (Psalms 90:4). Since one of our years is *365 1/4* days, a year on high is 365,250 of our years. Two years on high is 730,000 of our years. From this, continue multiplying to 49,000 years, each year consisting of *365 1/4* days, and each supernal day being one thousand of our years, as it is written, "God alone will prevail on that day" (Isaiah 2:11). "Who can speak of God's greatness?" (Psalms 106:2). Blessed be the name of Him whose glorious Kingdom is forever and ever.
All this relates to what the Scripture states. However, no matter how many times this picture is doubled, even thousands upon thousands, it would not even be like a second to *[God]* However, with regard to the Infinite One, it is enough that He is called Infinite.